A Unified First-Principles Pipeline for Metabolic Reaction Free Energies

Automatic QM routing over the ModelSEED reaction space

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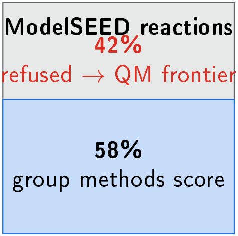
UMA machine-learned QM · pH-0 anion routing · self-gating · calibrated UQ

The problem: $\Delta_r G$ for the whole metabolic network

- Thermodynamics gates flux: every FBA / TFA model needs $\Delta_r G'^{\circ}$.
- Group-contribution (GCM, dGPredictor) is fast & accurate *in-domain* ...
- ... but **refuses ~42% of ModelSEED** — mass/charge-imbalanced, novel groups, off-distribution.
- **QM is parameter-free** and covers that frontier — *if* it can be made accurate and automatic.



20,802 balanced reactions are QM-runnable today (57× TECRDB).

Goal

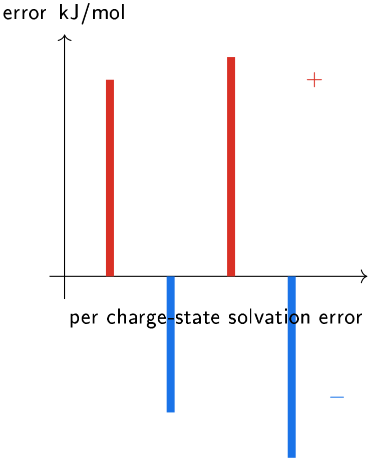
One pipeline, **no per-reaction tuning**, that scores any balanced reaction from structure alone.

Why naive QM fails on metabolites

Two failure modes, both large:

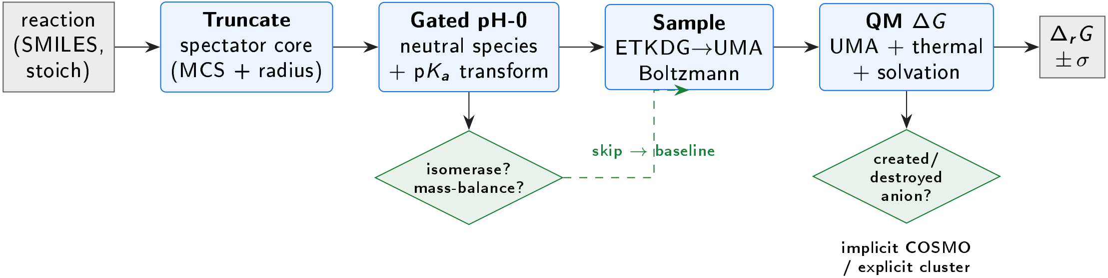
1. **Anion-solvation catastrophe.** Implicit continuum mis-solvates each polyphosphate charge state by $\pm$20–50 kJ/mol, and the errors are *sign-varying* — they do **not** cancel in $\Delta_r G$.
2. **Conformer noise** on huge, floppy substrates (sugar-phosphates, cofactors): spectator backbone dominates the energy and drowns the reactive signal.

- Worst for the ML potential (UMA): formal anionic charge + diffuse density + non-local delocalisation.
- Baseline MAE **29 kJ/mol**; *huge/floppy* class **45 kJ/mol**.



Opposite signs $\Rightarrow$ scatter, not a subtractable bias.

The idea: one self-routing pipeline



Every stage is a **general heuristic** — chosen from structure, no reaction-id, no fitting to the benchmark.

Stage 1 — Spectator truncation

- Map the reaction, keep the **reactive core** (atoms that change bonding) + a radius shell, cap open valences.
- Removes the conserved backbone → kills catastrophic cancellation *and* its conformer noise.
- **Radius-sensitivity guard**: escalate the shell until ΔG is stable (good cut $|\Delta\Delta G|$=1.3; bad cut 27.7 → rejected).
- v2 global-map extends it to the 20% multi-coefficient / unequal-side reactions.

Example

rxn03643 (glycosyl transfer)

full molecule: $-10\,kJ/mol$ error
truncated core: $+1.4\,kJ/mol$

After pH-0, truncated ≈ full-molecule error — truncation mainly buys *robustness* and speed.

Stage 2 — pH-0 neutral-species routing (the accuracy breakthrough)

Idea (Jinich 2014/18; Alberty transform): don't compute the mis-solvated anion. Protonate *every* anionic site to its **neutral** microspecies — UMA's comfortable regime — then bridge to pH 7 *analytically* with textbook pK_a's:

$$\Delta_r G'(\mathrm{pH}\,7) = \underbrace{\Delta_r G_{\mathrm{QM}}(\text{all-neutral})}_{\text{no formal charge}} + \sum_{\text{sites } i} s_i\, RT \ln\left(1+10^{\,\mathrm{pH}-\mathrm{p}K_{a,i}}\right), \quad s_i = \begin{cases} +1 & \text{react} \\ -1 & \text{prod} \end{cases}$$

- No atom-mapper: matched spectator anions **cancel** analytically.
- Max-anion canonicalisation → protonation-source-independent.
- Exact Alberty form (not linear) — correct near *and* above pK_a.
- Nothing fitted to the DB — pK_a's are textbook.

Validated

rxn00695 (adenylyltransferase)
$-96 \rightarrow +0.2\, kJ/mol$

rxn10427 (adenylate kinase)
$+61 \rightarrow +4.8\, kJ/mol$

Stage 2 makes itself safe: two structural gates

Isomerase gate

- If every reactant formula matches a product formula (a rearrangement), there is *no* anion-solvation change to fix.
- pH-0 would only inject neutral-vs-anion noise $\rightarrow$ **skip**, use baseline.

H-mass-balance guard

- pH-0 fixes $n_{\mathrm{H}^+}=0$ (the transform carries all proton exchange).
- Only valid when re-protonation is H-balanced:

$$\sum_i \nu_i H(\text{neutral}_i) = 0.$$

- Else refuse $\rightarrow$ baseline.

The bug this caught

Net-proton reactions (NAD(P)/GSH **redox**, deamination) drop one proton when forced to $n_{\mathrm{H}^+}=0$:

$$\approx 1170\ kJ/mol \text{ of garbage.}$$

21 such reactions gave $\pm 1150\ kJ/mol$ (e.g. rxn00184 $+1112$). The guard refuses **all 21** $\rightarrow$ they self-route to baseline; the validated phosphate class is untouched. **Zero accuracy cost.**

General & testable — works on ModelSEED with no reaction-id.

Stages 3–4 — Adaptive sampling & solvation triage

Convergent conformer sampling

- ETKDG pool $\rightarrow$ *batched* UMA single-point rank $\rightarrow$ relax lowest-k $\rightarrow$ Boltzmann ensemble (not min).
- Keep adding seed-batches until $G_{\mathbf{ens}}$ *and* min-E stop moving ($< 2.5\ kJ/mol$) — pool scales with rotatable bonds.
- Reports the batch-to-batch spread as sampling σ (feeds UQ).

Per-species solvation triage

- **Implicit** COSMO when no compact anion is created/destroyed (default).
- **Explicit** first-shell cluster-continuum (deterministic water count, Bryantsev monomer cycle) for a created/destroyed high-charge-density anion.
- Spectator-anion guard: explicit only where waters *cancel*.

Explicit *solves* the anion problem too

nucleotidyl transfer: implicit $-28..-52 \rightarrow +4\ kJ/mol$ (exp $+1.9$).

A validated *co-equal* anion fix. pH-0 is the default only on cost + noise ($\sigma \sim 1$–2 vs ~ 14 kJ run-to-run) + big-polyanion robustness — explicit stays the fallback where no clean pK_a ladder exists.

The QM backend & calibrated uncertainty

$$\Delta_r G = \underbrace{\Delta E_{\text{elec}}}_{\text{UMA gas}} + \underbrace{\Delta G_{\text{thermal}}}_{\text{UMA-Hessian RRHO}} + \underbrace{\Delta\Delta G_{\text{solv}}}_{\text{xtb continuum}} + n_{\mathrm{H}^+}\, G(\mathrm{H}^+_{\mathrm{aq}}, \mathrm{pH}\,7)$$

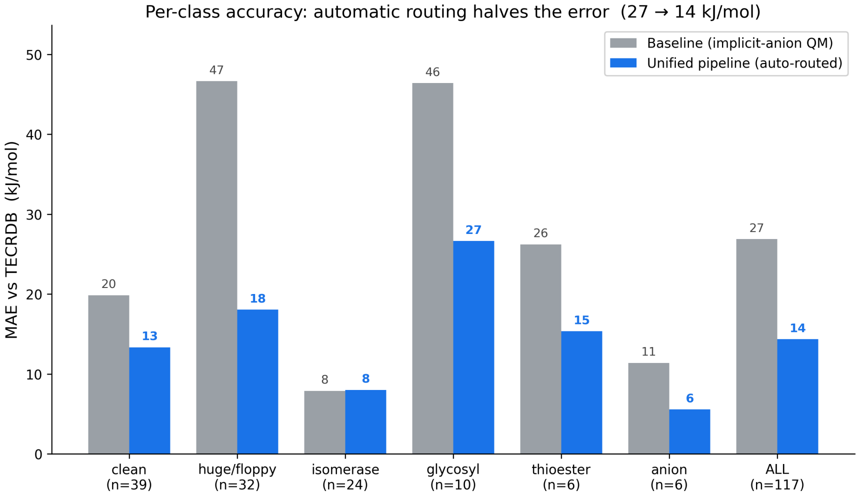
- Engine: **UMA** (`uma-s-1p2`, OMol25) — machine-learned QM, charge/spin aware.
- *Fast split*: UMA-Hessian thermal + xtb single-point solvation → ∼10× faster than `-ohess`, ΔG-neutral.
- Waters referenced to bulk liquid so counts cancel across the reaction.

Calibrated σ for downstream TFA/flux

$\sigma^2_{\text{total}} = \sigma^2_{\text{sampling}} + \sigma^2_{\text{class}} + \sigma^2_{\text{motif}}$

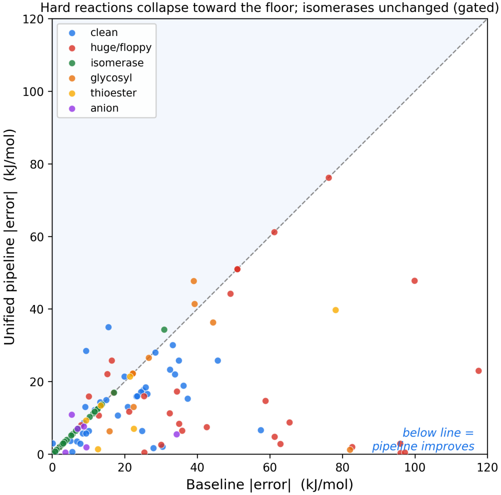
Cheap (lookups, no extra QM). Flags **UNRESOLVED** when $|\Delta G| < \sigma_{\text{samp}}$ (concentration-limited, not QM-fixable).

Result: automatic routing halves the error



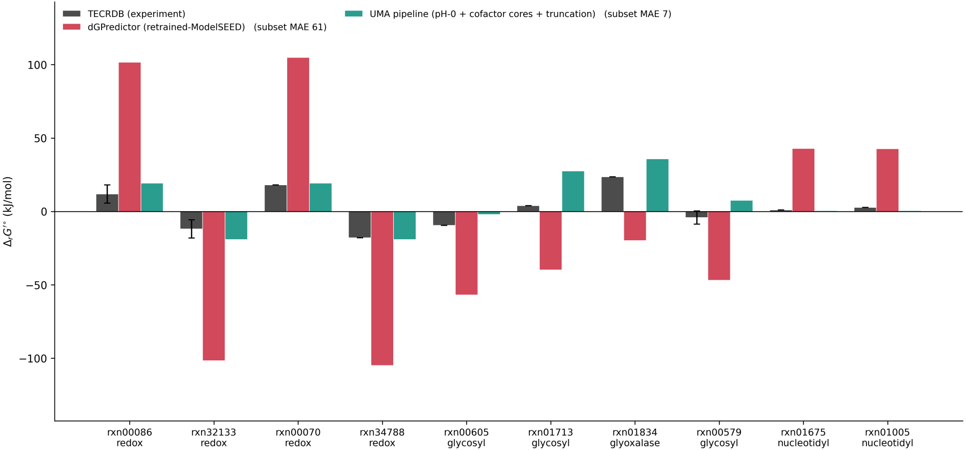
Matched subset, 117 processed so far. *huge/floppy* 47 → 18; *anion* (pure phosphoryl) 11 → 6 — pH-0 *solves* it; *glycosyl* (N-glycosidic transfer) 46 → 27 is the reference-method ceiling, not anion solvation; *isomerase* unchanged (gated).

Result: the high-error tail collapses



Points below the diagonal improve. The worst baseline reactions (80–118 kJ/mol) drop under 25 kJ/mol. Isomerases sit on the line — the gate leaves them alone.

Head-to-head where group-contribution struggles



The 10 reactions where retrained-dGPredictor disagrees most with experiment (cherry-picked, *not* representative). The QM pipeline (pH-0 + ring-cofactor + truncation) tracks experiment where the group method flies off by 80–100 kJ/mol — the value is **coverage of the frontier**, not beating dGPredictor on its average (full-367 dGP $\approx$ 5.7).

From 367 benchmark reactions to the database

- Validated on **TECRDB** (367 reactions with experimental $\Delta_r G$).
- Same code, one adapter → **20,802 ModelSEED reactions** runnable — no new flags.
- The router fires automatically: truncate → gated pH-0 → triage → QM → σ.
- `unified_pipeline.py` is the single entry point; everything else is a library.

Coherent auto-router

No manual per-reaction decisions. Each reaction picks its own truncation radius, pH-0 eligibility, solvation model, and uncertainty from structure alone.

Full-367 pH-0 sweep running across ∼37 GPUs; numbers here are the completed subset.

Honest open items

- **Residual is scatter, not bias** $\rightarrow$ calibration won't shrink it; *better referencing* will. Isodesmic / reference-reaction anchoring is the untapped multiplier.
- **Glycosyl electronic floor** is the *reference method*, not UMA (DFT $\approx$ UMA $+0.7\,kJ/mol$) — a genuine ceiling, flagged as reduced-confidence.
- **Mapping**: MCS vs RXNMapper — neither best alone (RXNMapper weak on phosphate, strong elsewhere); confidence-gated hybrid in progress.
- **Mg^{2+}**: subsumed by pH-0 on TECRDB; revisit only if a systematic-bias test demands explicit Mg.
- Finish the sweep $\rightarrow$ final aggregate + per-class σ calibration.

Summary

- A **single automatic pipeline** scores metabolic $\Delta_r G$ from structure — no per-reaction tuning.
- **pH-0 routing** collapses the anion-solvation catastrophe; two structural gates keep it safe (isomerase, mass-balance).
- Matched-subset MAE **27** $\rightarrow$ **14 kJ/mol**; *huge/floppy* **47** $\rightarrow$ **18**, pure-anion **11** $\rightarrow$ **6**.
- Scales to **20,802** ModelSEED reactions the group methods refuse.

Next

1. Finish sweep, calibrate σ.
2. Isodesmic referencing.
3. ModelSEED frontier demo (QM vs GCM).

Method: UMA + xtb + Alberty transform. Parameter-free.

Thank you — questions?